

Dipankar Saha

Senior AI / LLM Engineer

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Summary

Engineer building LLM-backed product features on solid distributed-systems foundations: multi-provider AI analysis pipelines, pgvector RAG (duplicate-incident detection, role-to-evidence retrieval), LangGraph agents behind a provider-ladder abstraction, and the event-driven microservices they run inside.

Skills

AI & LLM: OpenAI API, RAG (pgvector), LangGraph agents, Prompt Engineering, LLM Workflow Design

Backend: Node.js, NestJS, TypeScript, REST APIs, GraphQL, gRPC, JWT / OAuth2, RBAC

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Prisma, Drizzle ORM

Messaging & Architecture: RabbitMQ, Event-Driven Architecture, Microservices Design, Monorepo (pnpm / Turborepo)

Cloud & DevOps: AWS (EC2, S3, EKS, ECR, ELBv2), Kubernetes, Helm, ArgoCD (GitOps), Terraform, Docker, GitHub Actions, Jenkins, CI/CD

Observability: OpenTelemetry, Prometheus, Grafana, Jaeger, Loki

Frontend: React, Next.js (App Router), Tailwind CSS, Radix UI

Projects

Personal Growth AI OS

A private system that distills real engineering work into a proof-of-skills graph feeding career, content, and learning agents.

- Seven structured LLM agents behind a provider-ladder abstraction (Gemini / OpenAI / Anthropic) with automatic fallback; a LangGraph extraction agent. ([proof](#))
- A pgvector RAG knowledge base with cross-source duplicate detection and deterministic entity linking, feeding those agents. ([proof](#))
- A deterministic retrieval endpoint that maps a role's requirements to proof from shipped work — the same pipeline behind this résumé's own tailoring. ([proof](#))

Tech: Next.js, TypeScript, PostgreSQL, pgvector, Drizzle, LangGraph, OpenAI / Anthropic APIs, Vercel · [Live](#) · [GitHub](#)

AI-Powered Event-Driven Notification Platform

11-service NestJS platform that routes every event through tenant-scoped rules and an LLM analysis pass so only de-duplicated, severity-ranked notifications reach a person.

- Multi-provider AI event analysis and pgvector RAG duplicate-incident detection, running inside an 11-service NestJS platform (gRPC internally, REST at the edge) plus a Python/FastAPI prediction service. ([proof](#))
- Asynchronous RabbitMQ event choreography feeds the AI analysis pass on a single durable topic exchange; synchronous gRPC reserved for auth, tenancy, and template rendering. ([proof](#))
- Tenant-scoped OpenTelemetry → Prometheus / Loki / Jaeger / Grafana observability; k6 + HPA + KEDA load and autoscaling harness; Stripe billing. ([proof](#))

Tech: NestJS, TypeScript, gRPC, RabbitMQ, PostgreSQL, Redis, pgvector, OpenAI, FastAPI, Docker, Kubernetes, OpenTelemetry · [GitHub](#)

Platform Infrastructure/Platform GitOps — Terraform provision + Helm + ArgoCD Delivery

Terraform-provisioned AWS-shaped runtime plus the Helm + ArgoCD GitOps that deploys onto it — reproducible infra and reproducible delivery for a 13-service platform.

- Root Terraform module (~2000 lines) plus reusable modules/{network, eks, ecr, addons, loadbalancer} provisions a full AWS-shaped runtime (VPC, EKS, ECR, ELBv2); one flag switches the whole graph between a local k3s emulator and real AWS. [\(proof\)](#)
- Serial, idempotent bring-up of two k3s-backed clusters with a containerd registry mirror, a cross-cluster Service/Endpoints bridge, and an ELBv2-emulated ALB. [\(proof\)](#)
- One shared Helm chart templates all 11 NestJS services, layered with per-service, per-environment values files; ArgoCD auto-syncs from git with a bounded RollingSync ApplicationSet (maxUpdate 1) so a multi-service release doesn't spike CPU. [\(proof\)](#)
- Database migrations run as an ArgoCD PreSync hook Job keyed by a schema hash; Jenkins (itself a Helm-managed workload) builds/pushes each service image and commits the tag bump back for ArgoCD to pick up. [\(proof\)](#)
- ArgoCD, Jenkins, and the Jaeger/OTel/Prometheus/Grafana stack run as pinned-IP Docker containers; shell helpers enforce single-flight Terraform and minimal-touch restart reconcile. [\(proof\)](#)

Tech: Terraform, Kubernetes (k3s), AWS (EKS/ECR/ELBv2), Helm, ArgoCD, GitOps, Jenkins, Docker · [Infra repo](#) · [GitOps repo](#)

Experience

Senior Software Engineer — Veztraa Solutions

Remote | Mar 2023 — Present

- Design and build enterprise SaaS applications with React, Next.js, TypeScript, Node.js and NestJS, following a microservices architecture.
- Lead architecture discussions for scalable, cloud-native services — modular backends with REST APIs and event-driven messaging.
- Implement secure authentication and authorization with JWT and role-based access control.
- Optimize MongoDB aggregation pipelines and MySQL queries through indexing and schema tuning.
- Containerize services with Docker, deploy to Kubernetes on AWS, and maintain CI/CD with GitHub Actions and Jenkins.
- Mentor junior engineers on service design, testing, and delivery.

Tech: React, Next.js, TypeScript, Node.js, NestJS, Docker, Kubernetes, MySQL, MongoDB, RabbitMQ, AWS, GitHub Actions, Jenkins

Software Engineer — Tata Consultancy Services (TCS)

Kolkata, India | Apr 2021 — Feb 2023

- Built enterprise insurance applications for Zurich Insurance with Angular and Node.js.
- Developed scalable REST APIs and frontend modules for high-volume business workflows.
- Worked with distributed teams in Agile/Scrum; resolved production issues while keeping availability high.
- Improved application performance and reduced production defects through proactive optimization.

Tech: Angular, Node.js, JavaScript, REST APIs, MySQL, Git

Software Engineer — Asparrowtech

Indore, India | Jun 2020 — Mar 2021

- Built responsive web applications with JavaScript frameworks and Node.js.
- Developed backend APIs and integrated third-party services; contributed to database design and enhancements.
- Collaborated directly with clients to deliver custom software.

Tech: JavaScript, Node.js, REST APIs

Education

B.Tech in Information Technology, Siliguri Institute of Technology — 2016 – 2020